

VII Lecture

Support Vector Machines

7.1. Introduction and motivation

As seen in Lecture 6, when classifying into one of two p -dimensional multivariate normal populations, the scores are either linear (when the same covariance matrices are used) or quadratic (when the covariance matrices are different). Even optimality for such simple classifiers could be shown due to the multivariate normality assumption. However, when the two populations are **not** multivariate normal, the situation is more difficult, the bounds between the populations may be more blurry and significantly more non-linear classification techniques may be necessary to achieve a good classification. Support vector machines (SVM) are an example of such non-linear statistical classification techniques. They usually achieve superior results in comparison to more traditional non-linear parametric classification techniques such as *logit analysis* or non-parametric techniques such as *neural networks*. Mathematically, when using SVM, we try to formulate the classification as an empirical risk minimization problem and to solve the problem under additional restrictions on the allowed (nonlinear) classifier functions.

7.2 Expected versus Empirical Risk minimization

Let Y be an "indicator" with values $+1$ and -1 that indicate if certain p dimensional observation belongs to one of two groups of interest. We want to find a "best" classifier in a class $\mathcal{F}$ of functions f . Each classifier function $f(x)$ is meant to deliver a value of $+1$ or -1 for a given observation vector x . To this end, we consider the *expected risk*

$$R(f) = \int \frac{1}{2} |f(x) - y| dF(x, y)$$

Since the joint distribution $F(x, y)$ is unknown in practice, we consider the *empirical risk* over a training set $(x_i, y_i), i = 1, 2, \dots, n$ of observations instead:

$$\hat{R}(f) = \frac{1}{n} \sum_{i=1}^n \frac{1}{2} |f(x_i) - y_i|$$

The loss in the risk's definition is the "zero-one loss" given by

$$L(x, y) = \frac{1}{2} |f(x) - y|$$

and, thanks to the chosen labels ± 1 for Y obviously has the values 0 (if classification is correct) and 1 (if classification is wrong).

Minimizing the empirical (instead of the unknown expected) risk means to find $f_n = \operatorname{argmin}_{f \in \mathcal{F}} \hat{R}(f)$ as an approximation to $f_{opt} = \operatorname{argmin}_{f \in \mathcal{F}} R(f)$. Generally speaking the two solutions f_n and f_{opt} do not coincide and without further assumptions may be quite different. However, thanks to some ground breaking work by **V. Vapnik** there are theoretical results which, loosely speaking, state that if $\mathcal{F}$ is not too large and $n \rightarrow \infty$, there is an upper bound on their difference with probability $(1 - \eta)$,:

$$R(f) \leq \hat{R}(f) + \phi\left(\frac{h}{n}, \frac{\log \eta}{n}\right).$$

The above inequality can be interpreted as stating that the test error is bounded from above by the sum of the training error and the complexity of the set of models under consideration. We can then try to minimize the bound from above and hope that in that way we keep under control to a minimum the (unknown) test error.

The function ϕ above is monotone increasing in h (at least for large enough sample sizes n .) Here h denotes the Vapnik-Chervonenkis **(VC) dimension** (i.e., a measure of the complexity of the class $\mathcal{F}$).

For a linear classification rule $f(x) = \text{sign}(x'w + a)$ with a p dimensional predictor x it is known that

$$\phi\left(\frac{h}{n}, \frac{\log \eta}{n}\right) = \sqrt{\frac{h(\log(\frac{2n}{h}) + 1) - \log(\frac{\eta}{4})}{n}}.$$

and that the VC dimension is $h = p + 1$.

You can now **directly check** that

$$\frac{\partial}{\partial h} \left[\frac{h(\log(\frac{2n}{h}) + 1) - \log(\frac{\eta}{4})}{n} \right] = \frac{1}{n} \log\left(\frac{2n}{h}\right) > 0$$

as long as $h < 2n$ which confirms the monotone increasing property stated above.

In general, the VC dimension of a given set of functions is equal to the maximal number of points that can be separated in all possible ways by that set of functions.

At first glance, the “more rich” the function class $\mathcal{F}$ the better the classification rule would be. Indeed you can construct a classifier that has zero classification error on the training set. However, this classifier will be too specialised for the given training set with no ability to generalize for other sets. Hence such a classifier would be undesirable.

“More rich” is tantamount to require bigger complexity of $\mathcal{F}$ or equivalently higher value of h (and therefore of ϕ). The term $\phi(\frac{h}{n}, \frac{\log \eta}{n})$ can be considered penalty for the excessive complexity of the classifier function. You can see directly that the derivative $\frac{\partial \phi(\frac{h}{n}, \frac{\log \eta}{n})}{\partial h} \geq 0$ if and only if $2n \geq h$. For large enough n this means that the function ϕ is increasing with the complexity of the model. Hence the sum of the two terms: $\hat{R}(f)$ (precision) and $\phi(\frac{h}{n}, \frac{\log \eta}{n})$ (complexity) represents the compromise between precision in the risk estimation and the complexity of the classifier. Therefore minimizing this sum is the sensible thing to do in order to perform “optimally”.

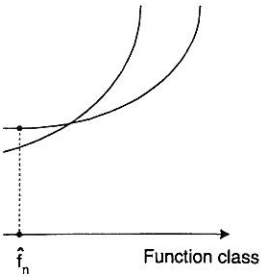
The rest of the lecture focusses on ways to solve (or solve approximately) this minimization problem for some classes $\mathcal{F}$.

It contains hard-copy extracts from Section 19.4 of

Härdle, W. and Simar, L., *Applied Multivariate Statistical Analysis*, Third Edition, Springer, 2012.

More detailed discussions and examples can also be found there. A treatment along similar lines can also be found in the e-book (available from the library)

Hastie, T., Friedman, J. and Tibshirani, R. *The Elements of Statistical Learning: Data Mining, Inference and Prediction*, Second Edition, Springer 2009.



and $\hat{R}(f)$ via the Vapnik-
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(19.18)

of the involved function
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(19.19)

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this is an easy condition

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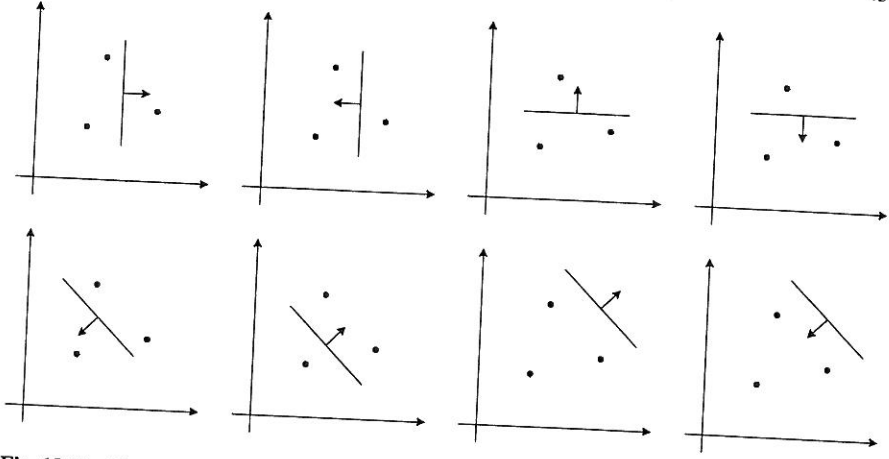
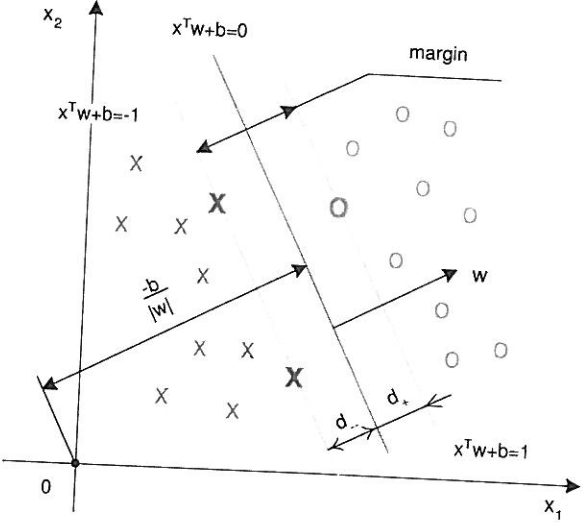


Fig. 19.11 Eight possible ways of shattering 3 points on the plane with a linear indicator function

Fig. 19.12 The separating
hyperplane $x^T w + b = 0$ and
the margin in the linearly
separable case



The SVM in the Linearly Separable Case

First we will describe the SVM in the linearly separable case. The family $\mathcal{F}$ of classification functions in the data space is given by:

$$\mathcal{F} = \{x^T w + b, w \in \mathbb{R}^p, b \in \mathbb{R}\} \quad (19.20)$$

In order to determine the support vectors we choose $f \in \mathcal{F}$ (or equivalently (w, b)) such that the so called margin – the corridor between the separating hyperplanes – is maximal. This situation is illustrated in Figure 19.12. The margin is equal to $d_- + d_+$. The classification function is a hyperplane plus the margin zone, where, in the separable case, no observations can lie. It separates the points from both classes

with the highest 'safest' distance (margin) between them. It can be shown that margin maximization corresponds to the reduction of complexity as given by the VC-dimension of the SVM classifier. Apparently, the separating hyperplane is defined only by the *support vectors* that hold the hyperplanes parallel to the separating one. In Figure 19.12 there are three support vectors that are marked with bold style: two crosses and one circle. We come now to the description of the SVM selection.

Let $x^T w + b = 0$ be a separating hyperplane. Then d_+ (d_-) will be the shortest distance to the closest objects from the classes $+1$ (-1). Since the separation can be done without errors, all observations $i = 1, 2, \dots, n$ must satisfy:

$$\begin{aligned} x_i^T w + b &\geq +1 & \text{for } y_i = +1 \\ x_i^T w + b &\leq -1 & \text{for } y_i = -1. \end{aligned}$$

We can combine both constraints into one:

$$y_i(x_i^T w + b) - 1 \geq 0, \quad i = 1, 2, \dots, n. \quad (19.21)$$

The *canonical hyperplanes* $x_i^T w + b = \pm 1$ are parallel and the distance between each of them and the separating hyperplane is $d_+ = d_- = 1/\|w\|$. To maximize the margin $d_+ + d_- = 2/\|w\|$ one therefore minimizes the Euclidean norm $\|w\|$ or its square $\|w\|^2$.

The Lagrangian for the primal problem that corresponds to margin maximization subject to constraint (19.21) is:

$$L_P(w, b) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i \{y_i(x_i^T w + b) - 1\}. \quad (19.22)$$

The Karush-Kuhn-Tucker (KKT) (Gale, Kuhn and Tucker, 1951) first order optimality conditions are:

$$\begin{aligned} \frac{\partial L_P}{\partial w_k} = 0: \quad & w_k - \sum_{i=1}^n \alpha_i y_i x_{ik} = 0, \quad k = 1, \dots, d \\ \frac{\partial L_P}{\partial b} = 0: \quad & \sum_{i=1}^n \alpha_i y_i = 0 \\ & y_i(x_i^T w + b) - 1 \geq 0, \quad i = 1, \dots, n \\ & \alpha_i \geq 0 \\ & \alpha_i \{y_i(x_i^T w + b) - 1\} = 0. \end{aligned}$$

From these first order condition, we can derive $w = \sum_{i=1}^n \alpha_i y_i x_i$ and therefore the summands in (19.22) read:

$$\begin{aligned} \frac{1}{2} \|w\|^2 &= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j \\ - \sum_{i=1}^n \alpha_i \{y_i(x_i^T w + b) - 1\} &= - \sum_{i=1}^n \alpha_i y_i x_i^T \sum_{j=1}^n \alpha_j y_j x_j + \sum_{i=1}^n \alpha_i \\ &= - \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j + \sum_{i=1}^n \alpha_i. \end{aligned}$$

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$\sum_{i=1}^n \alpha_i y_i x_i$ and therefore

$${}_j x_i^T x_j$$

$$x_j y_j x_j + \sum_{i=1}^n \alpha_i$$

$${}_j x_i^T x_j + \sum_{i=1}^n \alpha_i.$$

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Substituting this into (19.22) we obtain the Lagrangian for the dual problem:

$$L_D(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j. \quad (19.23)$$

The primal and dual problems are:

$$\min_{w,b} L_P(w, b)$$

$$\max_{\alpha} L_D(\alpha) \quad \text{s.t.} \quad \alpha_i \geq 0, \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

Since the optimization problem is convex the dual and primal formulations give the same solution.

Those points i for which the equation $y_i(x_i^T w + b) = 1$ holds are called support vectors. After "training the support vector machine" i.e. solving the dual problem above and deriving Lagrange multipliers (they are equal to 0 for non-support vectors) one can classify a company. One uses the classification rule:

$$g(x) = \text{sign}(x^T w + b), \quad (19.24)$$

where $w = \sum_{i=1}^n \alpha_i y_i x_i$ and $b = \frac{1}{2}(x_{+1} + x_{-1})^T w$. x_{+1} and x_{-1} are two support vectors belonging to different classes for which $y(x^T w + b) = 1$. The value of the classification function (the score of a company) can be computed as

$$f(x) = x^T w + b. \quad (19.25)$$

Each score $f(x)$ uniquely corresponds to a default probability (PD). The higher $f(x)$ the higher the PD.

SVMs in the Linearly Non-separable Case

In the linearly non-separable case the situation is like in Figure 19.13. The slack variables ξ_i represent the violation from strict separation. In this case the following inequalities can be induced from Figure 19.13:

$$\begin{aligned} x_i^T w + b &\geq 1 - \xi_i & \text{for } y_i = 1, \\ x_i^T w + b &\leq -1 + \xi_i & \text{for } y_i = -1, \\ \xi_i &\geq 0. \end{aligned}$$

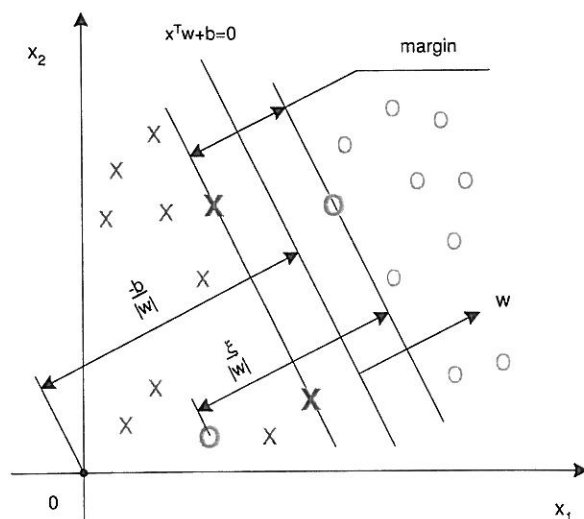
They can be combined into two constraints:

$$y_i(x_i^T w + b) \geq 1 - \xi_i \quad (19.26)$$

$$\xi_i \geq 0. \quad (19.27)$$

SVM classification again maximizes the margin given a family of classification functions $\mathcal{F}$.

Fig. 19.13 The separating hyperplane $x^T w + b = 0$ and the margin in the linearly non-separable case



The penalty for misclassification, the classification error $\xi_i \geq 0$, is related to the distance from a misclassified point x_i to the canonical hyperplane bounding its class. If $\xi_i > 0$, an error in separating the two sets occurs. The objective function corresponding to penalized margin maximization is then formulated as:

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i, \quad (19.28)$$

where the parameter C characterizes the weight given to the classification errors. The minimization of the objective function with constraint (19.26) and (19.27) provides the highest possible margin in the case when classification errors are inevitable due to the linearity of the separating hyperplane. Under such a formulation the problem is convex.

The Lagrange function for the primal problem is:

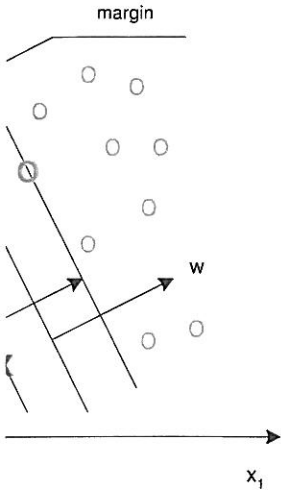
$$L_P(w, b, \xi) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i - \sum_{i=1}^n \alpha_i \{y_i(x_i^T w + b) - 1 + \xi_i\} - \sum_{i=1}^n \mu_i \xi_i, \quad (19.29)$$

where $\alpha_i \geq 0$ and $\mu_i \geq 0$ are Lagrange multipliers. The primal problem is formulated as:

$$\min_{w, b, \xi} L_P(w, b, \xi).$$

The first order conditions in this case are:

$$\frac{\partial L_P}{\partial w_k} = 0: \quad w_k - \sum_{i=1}^n \alpha_i y_i x_{ik} = 0$$



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$$b) - 1 + \xi_i) - \sum_{i=1}^n \mu_i \xi_i, \quad (19.29)$$

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$$\frac{\partial L_P}{\partial b} = 0: \sum_{i=1}^n \alpha_i y_i = 0$$

$$\frac{\partial L_P}{\partial \xi_i} = 0: C - \alpha_i - \mu_i = 0.$$

With the conditions for the Lagrange multipliers:

$$\alpha_i \geq 0$$

$$\mu_i \geq 0$$

$$\alpha_i \{y_i (x_i^T w + b) - 1 + \xi_i\} = 0$$

$$\mu_i \xi_i = 0.$$

Note that $\sum_{i=1}^n \alpha_i y_i b = 0$ therefore similar to the linear separable case the primal problem translates into:

$$\begin{aligned} L_D(\alpha) &= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j - \sum_{i=1}^n \alpha_i y_i x_i^T \sum_{j=1}^n \alpha_j y_j x_j \\ &\quad + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i - \sum_{i=1}^n \alpha_i \xi_i - \sum_{i=1}^n \mu_i \xi_i \\ &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j + \sum_{i=1}^n \xi_i (C - \alpha_i - \mu_i). \end{aligned}$$

Since the last term is 0 we derive the dual problem as:

$$L_D(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j x_i^T x_j, \quad (19.30)$$

and the dual problem is posed as:

$$\max_{\alpha} L_D(\alpha),$$

subject to:

$$0 \leq \alpha_i \leq C,$$

$$\sum_{i=1}^n \alpha_i y_i = 0.$$

Nonlinear Classification

The SVMs can also be generalized to the nonlinear case. In order to obtain non-linear classifiers as in Figure 19.14 one maps the data with a non-linear structure via a function $\Psi: \mathbb{R}^p \mapsto \mathbb{H}$ into a very large dimensional space $\mathbb{H}$ where the classification rule is (almost) linear. Note that all the training vectors x_i appear in L_D

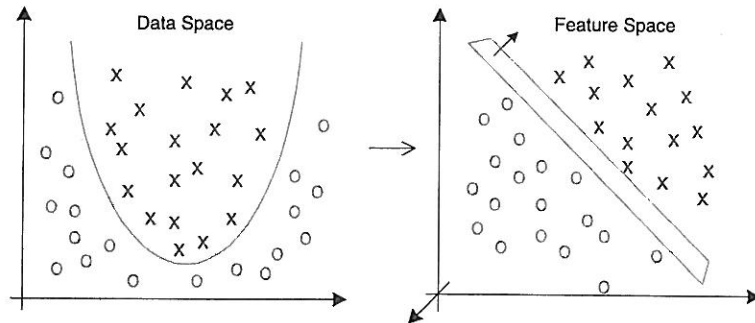


Fig. 19.14 Mapping into a three dimensional feature space from a two dimensional data space $\mathbb{R}^2 \mapsto \mathbb{R}^3$. The transformation $\Psi(x_1, x_2) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)^T$ corresponds to the kernel function $K(x_i, x_j) = (x_i^T x_j)^2$

(19.30) only as scalar products of the form $x_i^T x_j$. In the nonlinear SVM situations this transforms to $\psi(x_i)^T \psi(x_j)$.

The so called *kernel trick* is to compute this scalar product via a kernel function. These kernel functions are actually related to those we presented in Section 1.3. If a kernel function K exists such that $K(x_i, x_j) = \Psi(x_i)^T \Psi(x_j)$, then it can be used without knowing the transformation Ψ explicitly. A necessary and sufficient condition for a symmetric function $K(x_i, x_j)$ to be a kernel is given by Mercer's theorem (Mercer, 1909). It requires positive definiteness, i.e. for any data set $x_1, \dots, x_n$ and any real numbers $\lambda_1, \dots, \lambda_n$ the function K must satisfy

$$\sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j K(x_i, x_j) \geq 0. \quad (19.31)$$

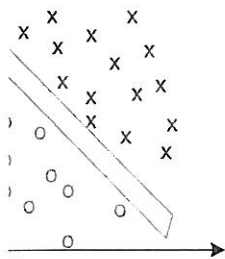
Some examples of kernel functions are:

- $K(x_i, x_j) = e^{-\|x_i - x_j\|/2\sigma^2}$ – the isotropic Gaussian kernel with constant σ
- $K(x_i, x_j) = e^{-(x_i - x_j)^T r^{-2} \Sigma^{-1} (x_i - x_j)/2}$ – the stationary Gaussian kernel with an anisotropic radial basis with constant r and variance-covariance matrix Σ from training set
- $K(x_i, x_j) = (x_i^T x_j + 1)^p$ – the polynomial kernel of degree p
- $K(x_i, x_j) = \tanh(kx_i^T x_j - \delta)$ – the hyperbolic tangent kernel with constant k and δ .

SVMs for Simulated Data

The basic parameters of SVMs are on the scaling r of the anisotropic radial basis functions (in the stationary Gaussian kernel) and the capacity C . The parameter r controls the local resolution of the SVM in the sense that smaller r create smaller curvature of the margin. The capacity C controls the amount of slack to allow for

Feature Space



a two dimensional data space responds to the kernel function

nonlinear SVM situations

duct via a kernel function. presented in Section 1.3. If $\Psi(x_j)$, then it can be used ssary and sufficient condi- given by Mercer's theorem any data set $x_1, \dots, x_n$ and

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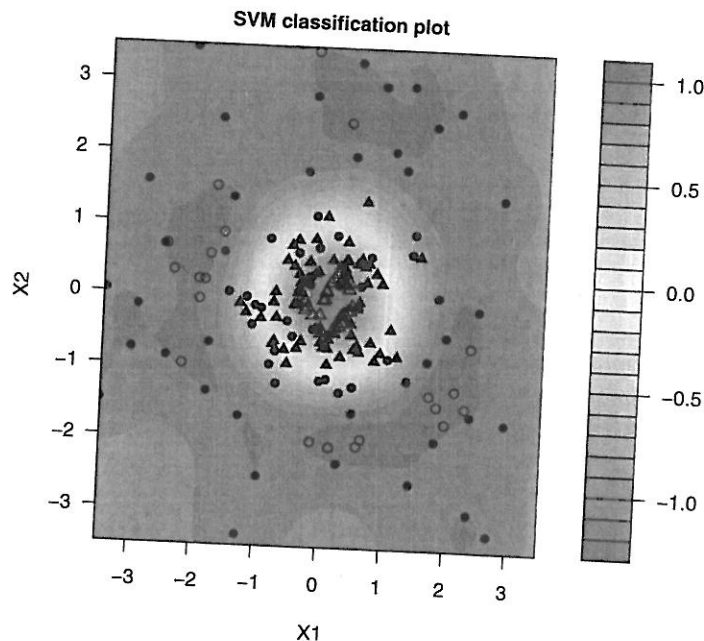


Fig. 19.15 SVM classification results for the "orange peel" data, $n = 200$, $d = 2$, $n_{-1} = n_{+1} = 100$, $x_{+1,i} \sim N((0, 0)^T, 2^2 \mathcal{I})$, $x_{-1,i} \sim N((0, 0)^T, 0.5^2 \mathcal{I})$ with SVM parameters $r = 0.5$ and $C = 20/200$ mva3svm01

unclassified observations. A large C would create a very rough and curved margin where C close to zero makes the margin more smooth.

One of the guinea pig tests for a classification algorithm is the data described as "orange peel", i.e. when two groups of observations have similar means, their variance, however, being different. The classification results in this case are presented in Figure 19.15. An SVM with a radial basis kernel is highly suitable for such a kind of data.

Another popular non-linear test is the classification of "spiral data". We generated two spirals with the distance between them equal 1.0 that span over 3π radian. The SVM was chosen with $r = 0.1$ and $C = 10/n$. The SVM was able to separate the classes without an error if noise with parameters $\varepsilon_i \sim N(0, 0.1^2 \mathcal{I})$ was injected into the pure spiral data (Figure 19.16). Obviously, both the "orange peel" and the "spiral data" are not linearly separable.

Solution of the SVM Classification Problem

The standard SVM optimization problem (19.30), which is a quadratic optimization problem, is usually solved by means of quadratic programming (QP). This technique, however, is notorious for (i) its bad scaling properties (the time required to

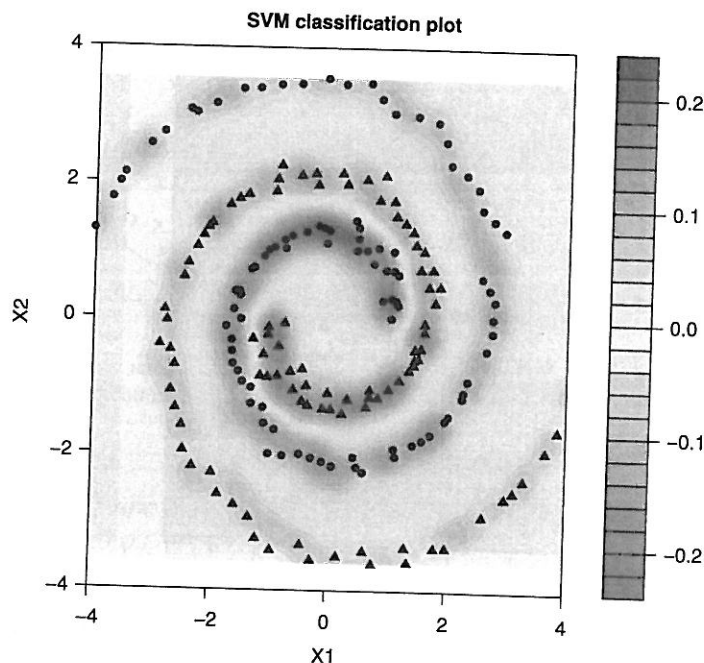


Fig. 19.16 SVM classification results for the noisy spiral data. The spirals spread over 3π radian; the distance between the spirals equals 1.0. $d = 2$, $n_{-1} = n_{+1} = 100$, $n = 200$. The noise was injected with the parameters $\epsilon_i \sim N(0, 0.1^2 I)$. The separation is perfect with SVM parameters $r = 0.1$ and $C = 10/200$ Q mva3svm02

solve the problem is proportional to n^3 , where n is the number of observations), (ii) implementation difficulty and (iii) enormous memory requirements. With the QP technique the whole kernel matrix of the size $n \times n$ has to be fit in the memory, which, assuming that each variable takes up 10 bytes of memory, will require $10 \times n \times n$ bytes. This means that 1 million observation (which is not unusual for practical applications such as credit scoring) will require 12000 TBytes (terabytes) or 10000000 MBytes of operating memory to store. With a typical size of the computer memory of 512 MBytes no more than around 5000 observations can be processed. Thus, the main emphasis in designing new algorithms was made on using special properties of SVMs to speed up the solution and reduce memory requirements.

Scoring Companies

For our illustration we selected the largest bankrupt companies with the capitalization of no less than 1 billion USD. The dataset used in this work is from the Credit reform database provided by the Research Data Center (RDC) of the Humboldt Universität zu Berlin. It contains financial information from about 20000 solvent and